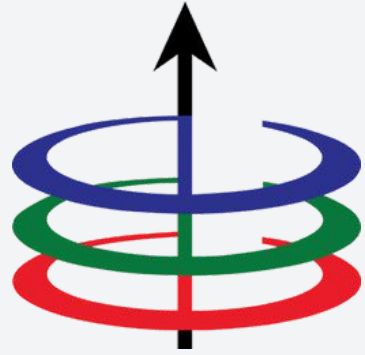




Lecture 8

Disjoint Set Data Structure (Union-Find)

EEE 121 THVW1

Steven S. Sison

- Disjoint Set DS (Union-Find)
- Implementation
- Applications

- **Disjoint Set DS (Union-Find)**
- Implementation
- Applications

Disjoint Sets

A disjoint set data structure, also known as Union-Find, is a collection of sets $S = \{S_1, S_2, S_3, \dots, S_N\}$ that is disjoint and dynamic.

- A given subset $\mathbf{S_A}$ and $\mathbf{S_B}$ is disjoint if $\mathbf{S_A} \cap \mathbf{S_B} = \emptyset$.
- The dynamic property means the contents of a set S can change.
- Each set is identified by some member of the set called the **representative**.

Operations:

1. MakeSet(N): creates N sets where each set consists of only one element.
2. Union(x, y): unites the sets that contain the elements x and y into a new set. The sets of x and y can then be removed.
3. FindSet(x): returns the representative of the unique set containing element x .

Why do we need Disjoint Sets?

Disjoint Set data structures are particularly useful in applications where you need to:

- Determine if two elements belong to the same set (Find)
- Combine two subsets into one single subset (Union)

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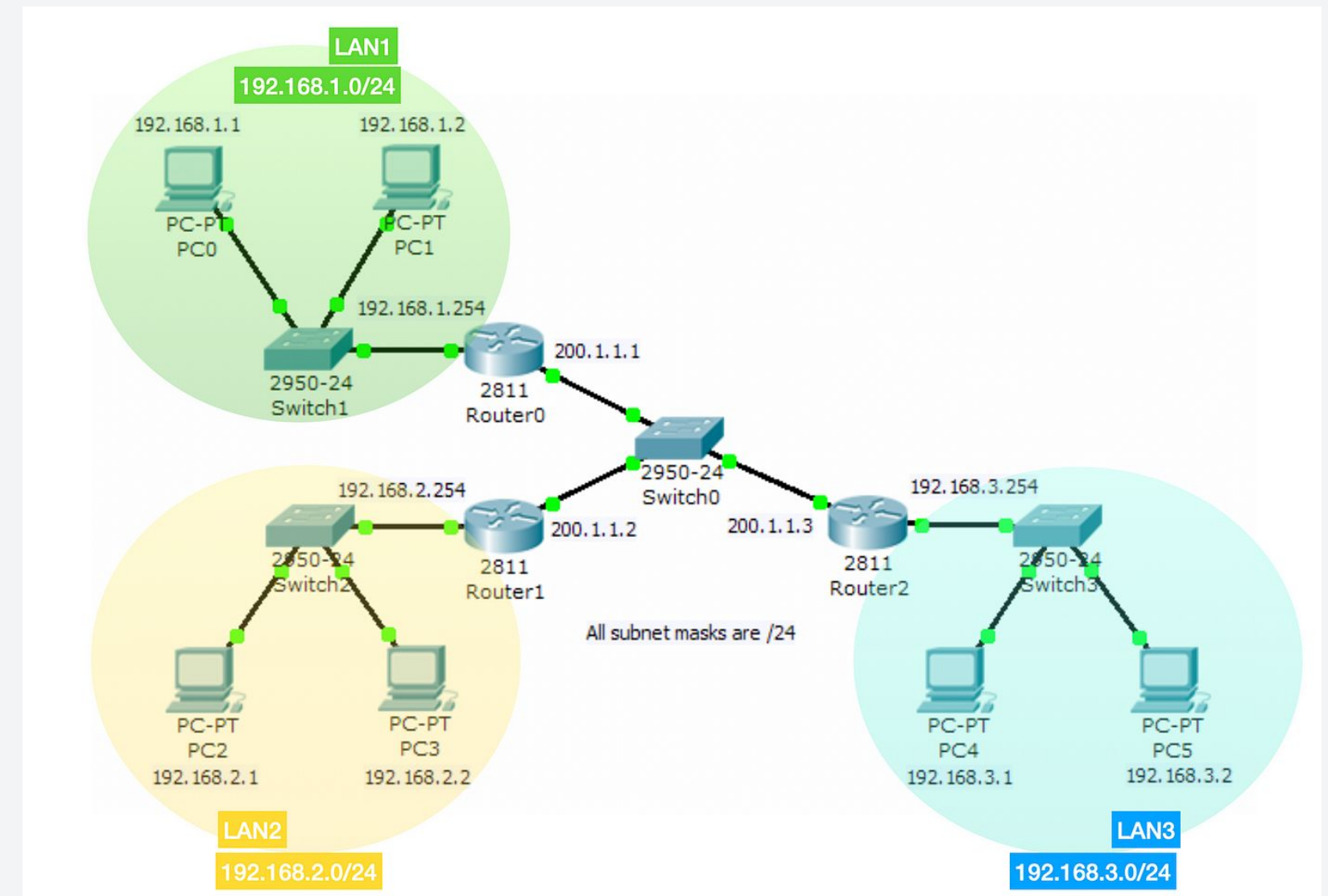
- Determine if two elements belong to the same set (Find)
- Combine two subsets into one single subset (Union)

Where do we see this?

- **Internet and Computer Networks:**

Using disjoint sets, we can determine if two devices are from the same network (subset).

- This information helps us to optimize routing protocols to these devices.



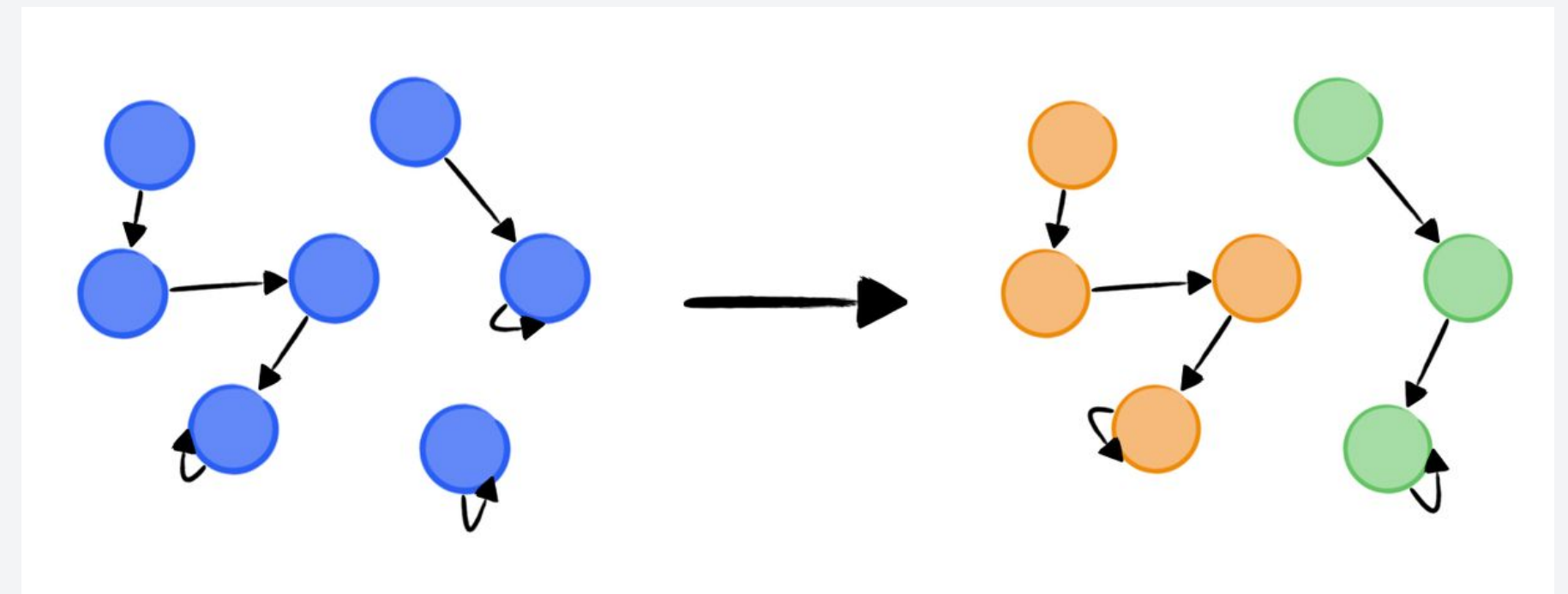
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- Determine if two elements belong to the same set (Find)
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Where do we see this?

- **Road Networks:** Using disjoint sets, we can use the Find operation to determine if two cities are connected by a road or not. Additionally, we can use the Union operation to combine two disjoint road networks.



- Disjoint Set DS (Union-Find)
- **Implementation**
- Applications

How do we implement Disjoint Sets?

Intuitively, we can implement disjoint sets using two arrays – the first array contains the elements while the second array contains the representative that the element belongs to.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
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In this example:

- **Element 1, 2, 4, 8, and 16 are the representatives.**
- This means that element 3 belongs to set 1, element 10 belongs to set 2, so on and so forth.

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In this example:

- To find the set of an element x , we use $\text{FindSet}(x)$.
 - $\text{FindSet}(3) = 1$
 - $\text{FindSet}(10) = 2$
- **What do you think is the time complexity of $\text{FindSet}(x)$?**

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- But what about union?

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In this example:

- But what about union?
 - We need to traverse the whole array to find the elements of a set and change its representative.
- **It would take $O(N)$**

How do we implement Disjoint Sets?

So how do we speed it up? We can represent each disjoint set as a tree.

- The root of the tree acts as the representative of the set.

Let's try it with the previous example

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
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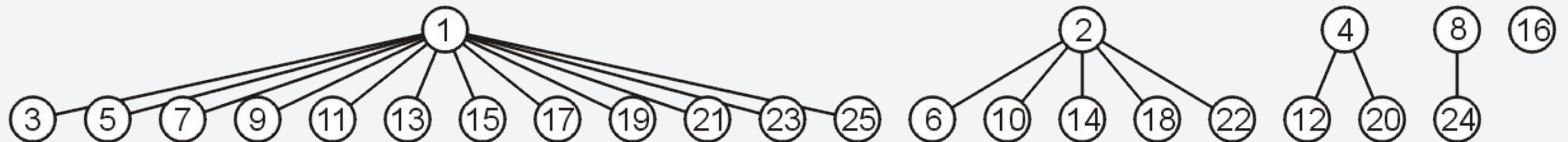
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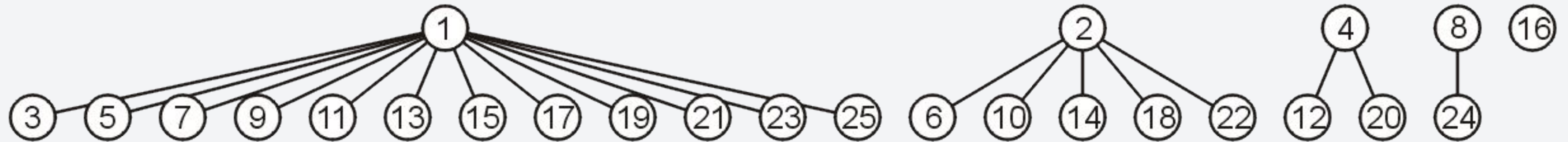
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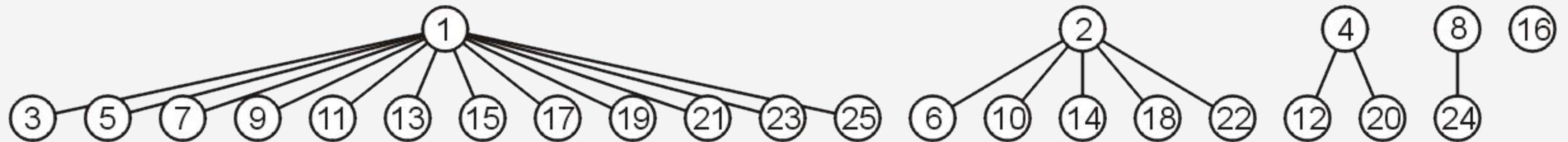


How is the union operation then performed?

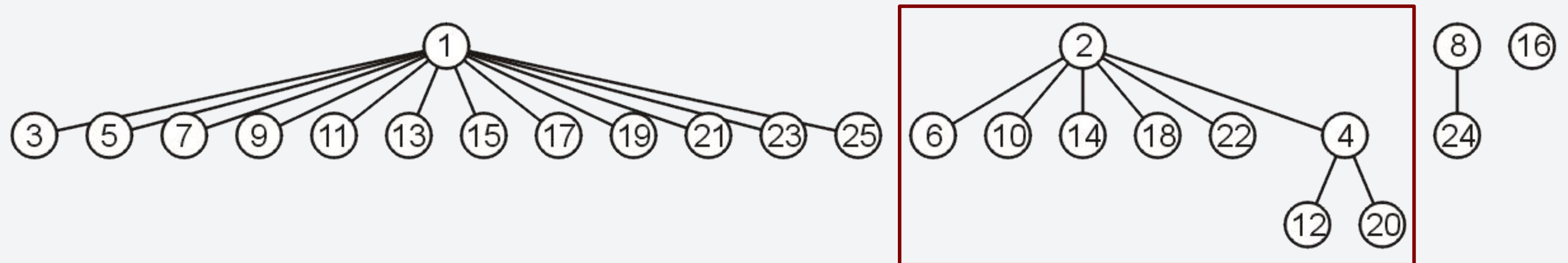
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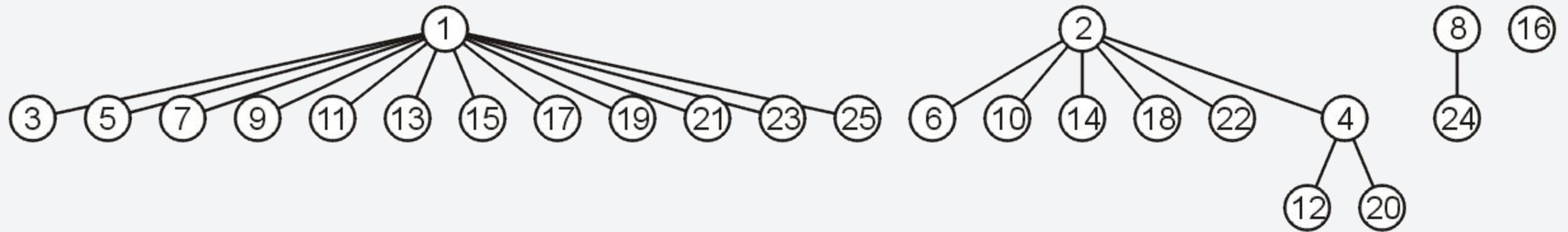


To take the union of two sets, we just need to attach the root of one tree to the root of another.



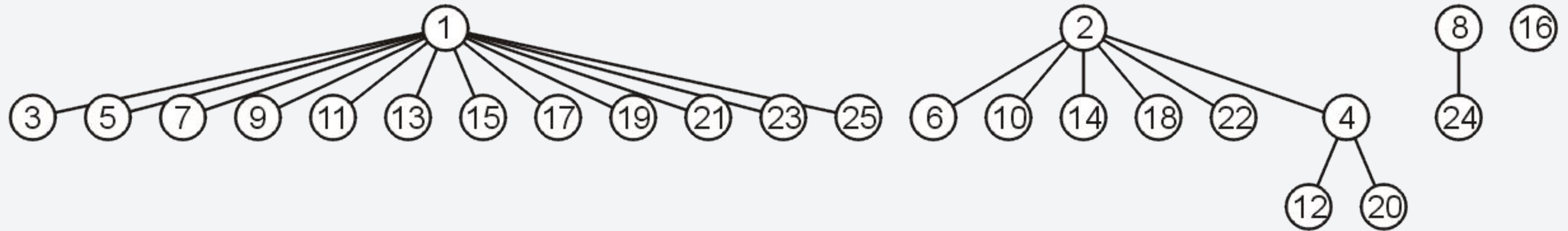
How do we implement Disjoint Sets?

Given these structure, can you determine the time complexity of find and union?



How do we implement Disjoint Sets?

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Both FindSet(x) and Union(x,y) have $O(h)$ time complexity where h is the height of the tree.

- FindSet(x): From the element, we need to traverse upwards to find the root.
- Union(x,y): From the element x and y , we need to find their root nodes separately making the time complexity $O(h_x + h_y) = O(h)$.

Code Implementation for Disjoint Sets

Initialization:

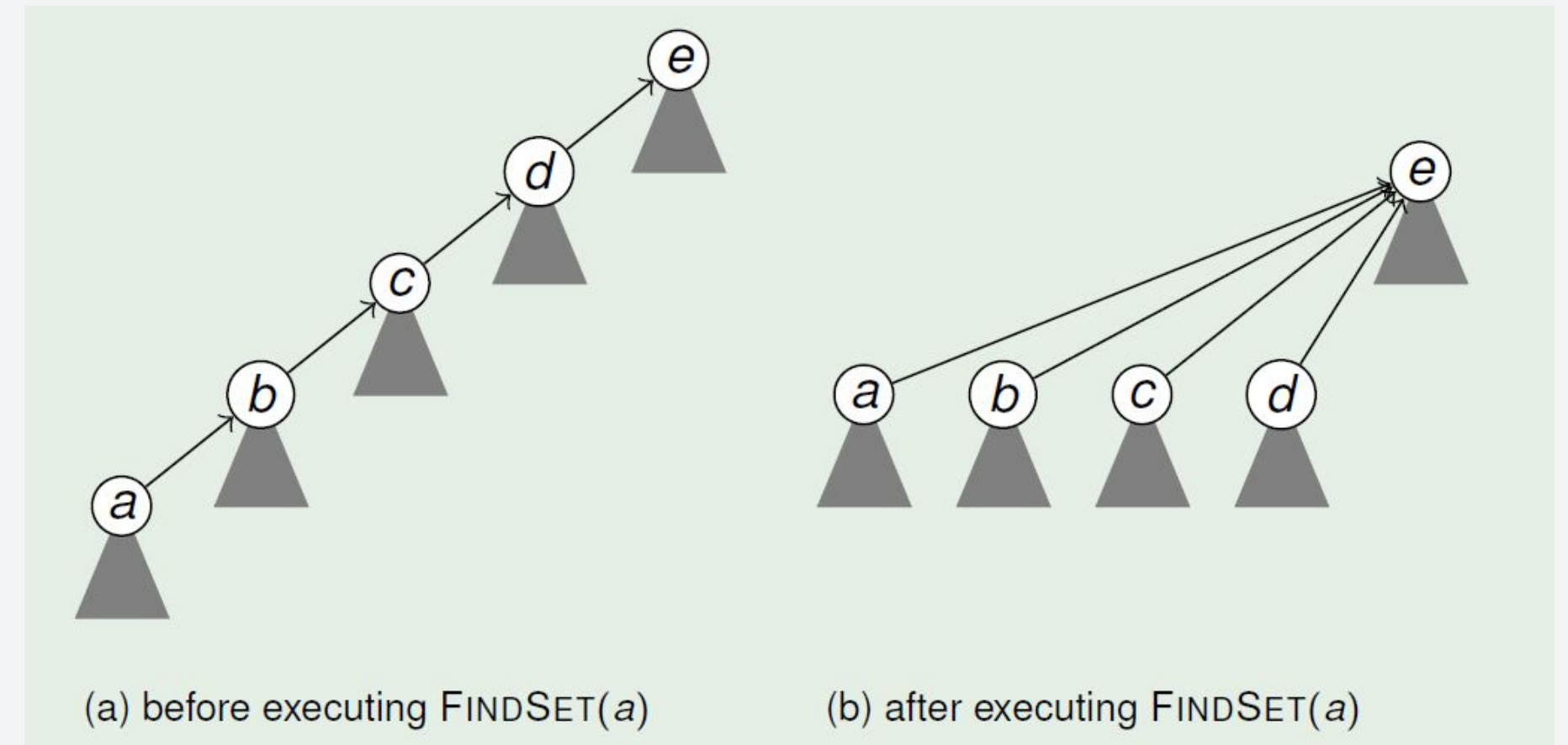
```
void MakeSet(int N) {  
    for(int i = 0; i < N; i++)  
        parent[i] = i; //representative of the set is itself  
}
```

- MakeSet(N) creates a parent array where each index is an element, and the value that the index contains is the representative of the element.
- To become a root node, $\text{parent}[i] == i$.

Code Implementation for Disjoint Sets

The FindSet(x) function:

```
int FindSet(int x) {  
    if(parent[x] != x) {  
        x = FindSet(parent[x]);  
        parent[x] = x;  
    }  
    return x;  
}
```

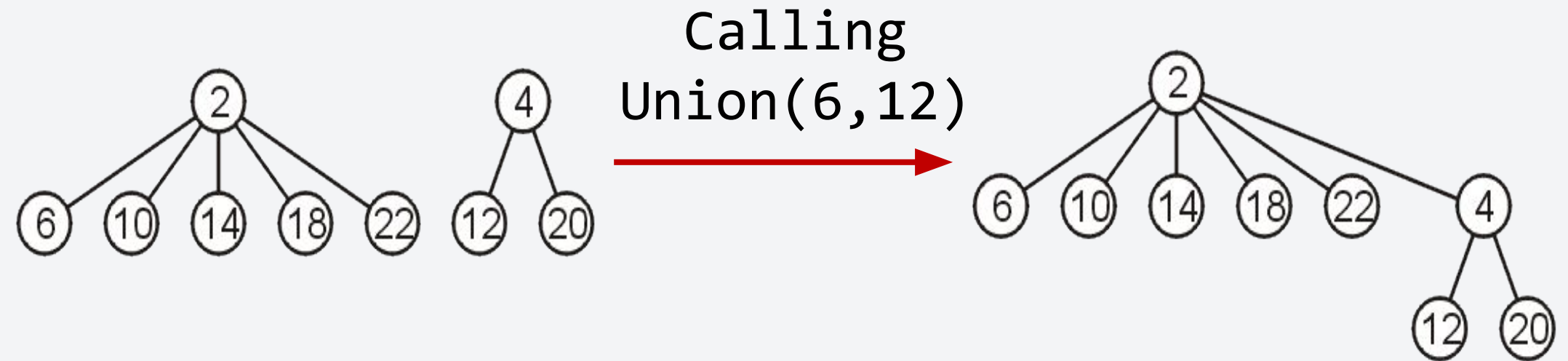


- The FindSet(x) operation is performed recursively to find the root node of a set.
- Additionally, it has a **path compression** built in wherein each node on the path will be made to point directly to the root node.
 - Recall: The union of two sets creates attaches one tree to another. The path compression will compress this tree as seen in the image.

Code Implementation for Disjoint Sets

The Union(x,y) function:

```
void Union(int x,int y) {  
    x = FindSet(x);  
    y = FindSet(y);  
  
    if(x != y)  
        parent[x] = y; //node y becomes the representative  
                        // of set where x belongs  
}
```



- The Union(x,y) function checks first if the elements x and y already have the same root node. If yes, the union is not performed. If no, then the union is performed.
- The Union(x,y) sets the **parent of the representative of x** to be equal to the representative of y. **NOTE: NOT JUST X BUT THE PARENT OF X.**

Code Implementation for Disjoint Sets

But what if the height of x is bigger than the height of y ?

- The resulting union would be a taller tree.
- We would need to change a lot of values during FindSet() due to **path compression**.

Code Implementation for Disjoint Sets

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Solution? We improve Union(x,y) by using a heuristic

- Main Idea: Make the root node of the tree with **smaller height** point to the root of the tree with bigger height.
- Our heuristic? Let's set a value **rank** and we perform **Union by Rank**:
 - Each node has a rank that approximates the logarithm of the size of the subtree at each node.
 - Serves as the upper bound of the height of the node.
 - The root with the smaller rank will be pointed to the root node with higher rank during Union(x,y)

Code Implementation for Disjoint Sets

Union(x,y) by Rank:

```
void Union(int x,int y) {  
    x = FindSet(x);  
    y = FindSet(y);  
  
    if(x != y) {  
        if(rank[x] > rank[y]) {  
            parent[y] = x;  
        } else if (rank[x] < rank[y]) {  
            parent[x] = y;  
        } else {  
            parent[x] = y;  
            rank[y] = rank[y] + 1;  
        }  
    }  
}
```

Note that we also update MakeSet() to include a new array – rank – where all elements are initialized to 0.

- Disjoint Set DS (Union-Find)
- Implementation
- **Applications**

Sample Application

Assumption: $G = (V, E)$ is an undirected graph.

1. Computing the connected components of G

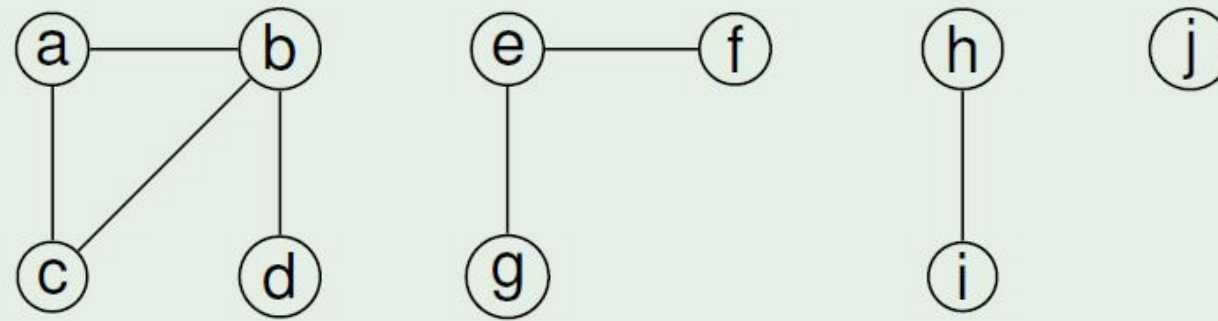
```
ConnectedComponents(G)
  MakeSet(V)
  for each edge  $(u, v) \in E$ 
    if FindSet(u)  $\neq$  FindSet(v)
      Union(u, v)
```

2. Determine if two elements are in the same component

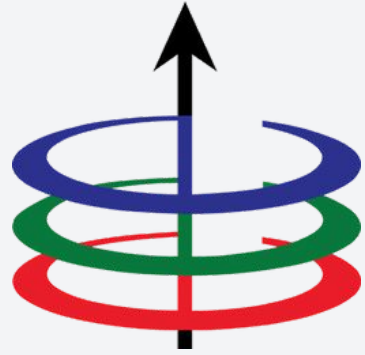
```
SameComponents(G)
  if FindSet(u) == FindSet(v)
    return True
  else
    return False
```

Sample Application

Example (A graph with 4 connected components)



Edge processed	Collection of disjoint sets									
initial sets	{a}	{b}	{c}	{d}	{e}	{f}	{g}	{h}	{i}	{j}
(b, d)	{a}	{b, d}	{c}		{e}	{f}	{g}	{h}	{i}	{j}
(e, g)	{a}	{b, d}	{c}		{e, g}	{f}		{h}	{i}	{j}
(a, c)	{a, c}	{b, d}			{e, g}	{f}		{h}	{i}	{j}
(h, i)	{a, c}	{b, d}			{e, g}	{f}		{h, i}		{j}
(a, b)	{a, b, c, d}				{e, g}	{f}		{h, i}		{j}
(e, f)	{a, b, c, d}				{e, f, g}			{h, i}		{j}
(b, c)	{a, b, c, d}				{e, f, g}			{h, i}		{j}



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